

Introduction to the Finite Element Method

Éric DARRIGRAND

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Introduction

Modelization and numerical simulation are nowadays essential tools of engineering sciences. Beside experimental approaches, the mathematical models and their resolution by numerical methods are widely used in many domains of physics, chemistry, biology or economy. Numerical methods aim to reduce the mathematical equations to finite dimensional problems such that the solutions to those equations are approximated by the solutions to finite dimensional systems. The numerical approaches are mandatory when the configuration is complicated such that there is no analytical solution of the mathematical model. The finite element method is one of these numerical techniques and was introduced in the 1950'. Firstly proposed by the physicists, the finite element method is now well known, carefully explained in a mathematical point of view, and implemented in many codes of numerical simulation. Several finite element libraries are also proposed on the web. In particular, the library MÉLINA++ ([11]) was developed at the university of Rennes 1, France, initially by Daniel Martin, former faculty member at the university of Rennes 1, now retired. MÉLINA++ is a generic finite element library which contains a wide range of finite elements and can solve numerous problems. The library is based on the consideration of the variational formulation and the involved integrands. A significant part of this document is inherited from a course by Daniel Martin on finite elements oriented towards the use of an initial version of MÉLINA++. Now, the library is maintained by Yvon Lafranche and Eric Darrigrand but is not anymore developed. The project of a new finite element library XLiFE++ ([5]) started a few years ago to extend the library MÉLINA++ to more complicated configurations requiring other families of finite elements.

The first chapter defines the general two-dimensional framework that we aim to consider in the document and introduces a fundamental ingredient for finite elements, the variational formulation. This presentation includes mathematical tools which are required for the justification of the variational formulation. The second chapter introduces the ingredients of the finite element methods in the framework of a two-dimensional model problem. This approach enables the reader to understand the main tools of the finite element methods in a rather simple configuration. Chapter 3 is devoted to a presentation of different families of finite elements. In Chapter 4, we detail the presentation of some technical ingredients of the finite element method: the elementary objects are carefully described; the assembling procedure is considered with storage concerns. These aspects are presented according to the choices done for the development of MÉLINA++. In the last chapter, some extensions to more complicated configurations are evoked such as three dimensional finite elements, time dependent problems, integral equations, Nédélec finite elements, or linear-system resolution issues.

Chapter 1

Motivation of Finite Elements

1.1 Elliptic problems – partial differential equations and eigenvalues problems

The generic configuration which we deal with in this document consists of elliptic partial differential equations or related eigenvalue problems. In this section, we first introduce very briefly these families of mathematical problems. Then, we define the generic boundary value problem which is considered in this document.

1.1.1 Elliptic boundary value problems – partial differential equations and eigenvalue problems

Elliptic boundary value problems are widely used for modelling physical phenomena. Depending on the problem, they can take several forms such as *partial differential equations* (PDE) or *eigenvalue problems*. In this document, we consider boundary value problems defined on a bounded domain Ω with boundary $\partial\Omega = \Gamma_D \cup \Gamma_N$.

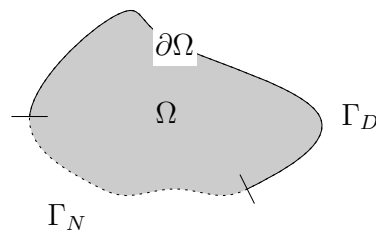


Figure 1.1: (generated with Fig4TeX [7]) A bounded 2D domain Ω

The *second-order elliptic partial differential equations* are defined thanks to an equation involving a second-order differential operator D . Combined with boundary conditions, they define a boundary value problem of the form:

$$\begin{cases} D u = f & \text{in } \Omega, \\ \text{boundary conditions on } \Gamma_N \text{ and } \Gamma_D. \end{cases}$$

The finite element method aims to approximate the solution to such a problem with a finite dimensional solution u_h assimilated to the vector solution U to a linear system of the form

$$D U = F ,$$

The system includes the consideration of the boundary conditions.

The *second-order elliptic eigenvalue problems* are defined thanks to an eigenvalue equation involving a second-order differential operator D . Combined with boundary conditions, they define a boundary value problem of the form:

$$\begin{cases} D u = \lambda u & \text{in } \Omega , \\ \text{boundary conditions on } \Gamma_N \text{ and } \Gamma_D . \end{cases}$$

where the unknown is the eigen pair (λ, u) . The finite element method aims to approximate the solution (λ, u) to such a problem with a finite dimensional solution (λ_h, u_h) where u_h is assimilated to the eigenvector U of a linear generalized eigenvalue problem

$$D U = \lambda_h M U ,$$

The eigenvalue system includes the consideration of the boundary conditions and the matrix M is the finite element approximation of the identity operator, known as the *mass matrix*.

1.1.2 A generic 2D elliptic partial differential equation

We consider an open, bounded, polygonal, convex domain $\Omega \subset \mathbb{R}^2$, of boundary $\partial\Omega = \Gamma_D \cup \Gamma_N$ such that $\Gamma_D \cap \Gamma_N$ is empty or reduced to a finite number of points of $\partial\Omega$. We denote by $x = (x_1, x_2)$ the generic point in Ω , by γ the generic point on $\partial\Omega$, and by $\nu(\gamma) = (\nu_1(\gamma), \nu_2(\gamma))$ the outgoing unit normal vector at point γ of $\partial\Omega$.

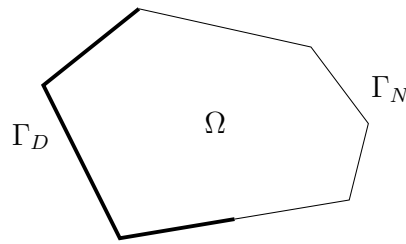


Figure 1.2: (generated with Fig4TeX [7]) The 2D domain Ω

We consider the functions

- ★ $a_{\alpha\beta}$ (for $\alpha, \beta = 1, 2$), a_{00} , f_Ω defined on Ω ,
- ★ b_N , g_N defined on Γ_N ,
- ★ u_D defined on Γ_D .

The generic PDE that we are going to deal with in this document is based on the differential operator defined by:

$$\begin{aligned} \mathcal{A}u(x) &= - \sum_{\alpha, \beta=1}^2 \frac{\partial}{\partial x_\beta} \left(a_{\alpha\beta}(x) \frac{\partial u}{\partial x_\alpha}(x) \right) + a_{00}(x) u(x), \quad x \in \Omega \\ \frac{\partial u}{\partial \nu_A}(\gamma) &= \sum_{\alpha, \beta=1}^2 a_{\alpha\beta}(\gamma) \frac{\partial u}{\partial x_\alpha}(\gamma) \nu_\beta(\gamma), \quad \gamma \in \partial\Omega. \end{aligned} \quad (1.1)$$

We then aim to explain the application of the finite element method in the context of the generic boundary value problem: find u solution to

$$\begin{cases} \mathcal{A}u = f_\Omega & \text{in } \Omega \\ \frac{\partial u}{\partial \nu_A} + b_N u = g_N & \text{on } \Gamma_N \quad \text{Neumann/Fourier condition} \\ u = u_D & \text{on } \Gamma_D \quad \text{Dirichlet condition} \end{cases} \quad (1.2)$$

A very simple case corresponds to the functions $a_{\alpha\beta} = \delta_\beta^\alpha$ ($\alpha, \beta = 1, 2$), where δ_β^α is the Kronecker delta, equal to 1 if $\alpha = \beta$ and 0 otherwise:

$$\mathcal{A}u = - \left(\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} \right) + a_{00} u = -\Delta u + a_{00} u \quad \text{and} \quad \frac{\partial u}{\partial \nu_A} = \text{grad } u \cdot \nu = \frac{\partial u}{\partial \nu}.$$

The consideration of these equations requires the introduction of suitable spaces, the so-called Sobolev spaces (see Section 1.2).

1.2 From PDE to variational formulation – Green’s formulae

A first step of the finite element method is the definition of a corresponding variational formulation. For mathematical justification, this task requires important tools of functional analysis and distribution theory such as Hilbert spaces, Sobolev spaces, trace operators, Green’s formulae. Numerous results are mentioned in this section with no demonstration. For more details or demonstrations, we refer the reader to literature (e.g. [13, 8, 10]).

1.2.1 Introduction to the functional spaces

The well known $\mathcal{C}^\alpha$ spaces are not sufficient for the mathematical justification of the different steps of the finite element method. We introduce here a specific family of Hilbert spaces which are crucial in the analysis of variational formulations. Some hypotheses are also required on the regularity of the domain. To this aim, we first give a definition for the regularity of Ω . Let $\mathcal{C}^{0,\alpha}(\Omega)$ be the set of functions f satisfying for all $(x, y) \in \Omega \times \Omega$:

$$\|f(x) - f(y)\| \leq \|x - y\|^\alpha$$

Definition 1.1. $\Omega \subset \mathbb{R}^d$ is said of class $\mathcal{C}^k$ (resp. $\mathcal{C}^{k,1}$) if for any point m_0 of the boundary $\partial\Omega$ of Ω , there exists a ball $B(m_0, R)$, with $R > 0$, a system of axes $\{O_0; x_1^0, \dots, x_d^0\}$ and a function $a_0 \in \mathcal{C}^k(\mathbb{R}^{d-1})$ (resp. $\mathcal{C}^{k,1}$) such that

$$\begin{aligned} \partial\Omega \cap B(m_0, R) &= \{x \in B(m_0, R) / x_d^0 = a_0(x_1^0, \dots, x_{d-1}^0)\} \\ \Omega \cap B(m_0, R) &= \{x \in B(m_0, R) / x_d^0 > a_0(x_1^0, \dots, x_{d-1}^0)\} \end{aligned}$$

Notions of Hilbert spaces

Definition 1.2 (norm, inner product). *Let H be a real space.*

★ *A norm on H is a function $\mathcal{N}$ defined from H to $\mathbb{R}^+$ satisfying:*

- ◇ $\forall u \in H, \forall \lambda \in \mathbb{R}, \quad \mathcal{N}(\lambda u) = |\lambda| \mathcal{N}(u),$
- ◇ $\mathcal{N}(u) = 0 \implies u = 0,$
- ◇ $\forall u, v \in H, \quad \mathcal{N}(u + v) \leq \mathcal{N}(u) + \mathcal{N}(v)$ (triangular inequality).

★ *We call inner product on H a function $\mathcal{P}$ defined from $H \times H$ to $\mathbb{R}$ satisfying:*

- ◇ $\mathcal{P}$ is bilinear symmetric,
- ◇ $\forall u \in H^*, \quad \mathcal{P}(u, u) > 0.$

★ *If $\mathcal{P}$ is an inner product on H , the function $\mathcal{N} : u \mapsto (\mathcal{P}(u, u))^{1/2}$ is a norm on H .*

Definition 1.3 (Cauchy sequence). *Let H be a normed space, with norm $\|\cdot\|$ and $(u_n)_{n \in \mathbb{N}}$ be a sequence of points of H .*

- ★ *$(u_n)_{n \in \mathbb{N}}$ is said convergent in $(H, \|\cdot\|)$ if there exists $u \in H$ / $\lim_{n \rightarrow \infty} \|u_n - u\| = 0$. We say that $(u_n)_{n \in \mathbb{N}}$ converges towards u in $(H, \|\cdot\|)$.*
- ★ *$(u_n)_{n \in \mathbb{N}}$ is said a Cauchy sequence if, for all $p \in \mathbb{N}$, $\lim_{n \rightarrow \infty} \|u_n - u_{n+p}\| = 0$.*
- ★ *A convergent sequence is a Cauchy sequence. The reciprocal is generally wrong.*
- ★ *If all Cauchy sequences are convergent, the space is said complete.*

Definition 1.4 (Hilbert space). *Let H be a real space.*

- ★ *H is said pre-Hilbert space if H is equipped with an inner product $\langle \cdot, \cdot \rangle$ and the corresponding norm $\|\cdot\|$.
Property (Cauchy-Schwarz inequality): $\forall u, v \in H, \quad |\langle u, v \rangle| \leq \|u\| \|v\|.$*
- ★ *H is said Hilbert space if H is a complete pre-Hilbert space.*

The spaces of integrable functions L^1 , L^p , equipped with their natural norms, are complete spaces ; they are Banach spaces. The space L^2 , equipped with its natural scalar product is a Hilbert space. The spaces C^α equipped with their natural scalar product are not Hilbert spaces. They suffer from a lack of completeness. Indeed, the Sobolev spaces that we are going to introduce can be interpreted as the completion of the corresponding spaces C^α equipped with the ad hoc scalar products.

Notions of distributions and dual product

In order to introduce the Sobolev spaces, we first briefly present the theory of distributions. A *distribution* is a concept which generalizes the notion of function and makes possible the operation of derivation in a particular sense so-called *distributional sense* or *weak sense*. In this section, we use the following notation: for $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d) \in \mathbb{N}^d$ a multi-index,

$$\partial^\alpha \varphi = \frac{\partial^{|\alpha|} \varphi}{\partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_d}^{\alpha_d}} \quad \text{with } |\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_d .$$

Definition 1.5 (test function space – distribution). *Let Ω be an open subset of $\mathbb{R}^d$.*

- ★ *A subset K of Ω is said compact if it is a closed and bounded subset of Ω .*
- ★ *We denote by $\mathcal{D}(\Omega)$ the set of functions which are C^∞ on $\mathbb{R}^d$ and the support of which is compact in Ω .*
- ★ *u is said a distribution on Ω if u is a linear form on $\mathcal{D}(\Omega)$*

$$u : \varphi \in \mathcal{D}(\Omega) \mapsto \langle u, \varphi \rangle$$

and satisfy the continuity property: for all compact K of Ω , there exists an integer k and a constant C_K such that

$$\forall \varphi \in \mathcal{D}(\Omega) \text{ such that } \text{supp}(\varphi) \subset K, \quad |\langle u, \varphi \rangle| \leq C_K \max_{|\alpha| \leq k} \left\{ \|\partial^\alpha \varphi\|_{L^\infty(\Omega)} \right\} .$$

- ★ *We denote by $\mathcal{D}'(\Omega)$ the space of distributions on Ω .*

In this definition, $\langle \cdot, \cdot \rangle$ is the duality product on $\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)$. For a function $f \in L^1_{\text{loc}}(\Omega) = \{f / f \in L^1(K), \forall K \text{ compact } \subset \Omega\}$, the linear form

$$\varphi \mapsto \int_{\Omega} f \varphi \, dx$$

defines a distribution. The continuity property is satisfied with $k = 0$ and $C_K = \int_K |f(x)| \, dx$. The distribution is identified to the function and we also denote by f the distribution associated with f .

The space of distributions is equipped with notions of convergence and derivation thanks to the following definition.

Definition 1.6 (distributional convergence and derivation). *Let Ω be an open subset of $\mathbb{R}^d$.*

- ★ *The sequence of distributions $(u_n)_{n \in \mathbb{N}}$ is said convergent towards $u \in \mathcal{D}'(\Omega)$ if and only if*

$$\forall \varphi \in \mathcal{D}(\Omega), \quad \langle u_n, \varphi \rangle \longrightarrow \langle u, \varphi \rangle \text{ in } \mathbb{R} .$$

★ For $u \in \mathcal{D}'(\Omega)$, $\Omega \subset \mathbb{R}^d$, we denote by $\frac{\partial u}{\partial x_i}$ ($1 \leq i \leq d$) the distribution defined by

$$\forall \varphi \in \mathcal{D}(\Omega), \quad \left\langle \frac{\partial u}{\partial x_i}, \varphi \right\rangle = - \left\langle u, \frac{\partial \varphi}{\partial x_i} \right\rangle .$$

If α is a multi-index, we denote $\partial^\alpha u$ the distribution defined by

$$\forall \varphi \in \mathcal{D}(\Omega), \quad \langle \partial^\alpha u, \varphi \rangle = (-1)^{|\alpha|} \langle u, \partial^\alpha \varphi \rangle .$$

Thanks to this definition, any function L^1 , L^p , which can be identified to a distribution, is derivable in the distributional sense.

Notions of Sobolev spaces

The previous definitions introduce a strong ingredient of functional analysis: the distributional derivation. When a function is classically derivable, its classical derivative coincides with the distributional derivative, but when the function is not derivable we consider the distributional derivative. In the sequel, we will consider either the classical or the distributional derivative without specifying.

Definition 1.7 (Sobolev space $H^1(\Omega)$). *Let Ω be an open subset of $\mathbb{R}^d$. We say that u is in $H^1(\Omega)$ if $u \in L^2(\Omega)$ and $\frac{\partial u}{\partial x_i} \in L^2(\Omega)$ for $i = 1, \dots, d$ (distributional derivatives ... of course). This space is equipped with the inner product*

$$\langle u, v \rangle_{H^1} = \int_{\Omega} u v \, dx + \sum_{i=1}^d \int_{\Omega} \frac{\partial u}{\partial x_i} \frac{\partial v}{\partial x_i} \, dx$$

and the corresponding norm

$$\|u\|_{H^1} = \left(\|u\|_{L^2}^2 + \sum_{i=1}^d \left\| \frac{\partial u}{\partial x_i} \right\|_{L^2}^2 \right)^{1/2}$$

Theorem 1.8. *The space $H^1(\Omega)$ equipped with this inner product $\langle \cdot, \cdot \rangle_{H^1}$ is a Hilbert space.*

Remark 1.9. *This definition can be extended:*

★ The spaces $H^m(\Omega)$, for $m \in \mathbb{N}$, defined by

$$H^m(\Omega) = \left\{ u \in L^2(\Omega) / \forall \alpha \text{ multi-index such that } |\alpha| \leq m, \partial^\alpha u \in L^2(\Omega) \right\} ,$$

equipped with the inner product

$$(u, v)_{H^m} = \sum_{|\alpha| \leq m} \int_{\Omega} \partial^\alpha u \partial^\alpha v \, dx ,$$

and the corresponding norm

$$\|u\|_{H^m} = \left(\sum_{|\alpha| \leq m} \|\partial^\alpha u\|_{L^2}^2 \right)^{1/2} ,$$

are Hilbert spaces.

★ When $\Omega = \mathbb{R}^d$, we define the spaces $H^s(\mathbb{R}^d)$ with $s \in \mathbb{R}^+$:

$$H^s(\mathbb{R}^d) = \{u \in L^2(\mathbb{R}^d) / (1 + |\xi|^2)^{s/2} \hat{u} \in L^2(\mathbb{R}^d)\} ,$$

where $\hat{u}$ is the Fourier transform of the function u . When s is an integer, the last definition is equivalent to the previous. The demonstration is based on Fourier transform of functions of L^2 and its distributional derivatives.

All these spaces satisfy properties such as density results or compactness injection. A crucial result concerns the completion $H_0^1(\Omega)$ of $\mathcal{D}(\Omega)$ in $(H^1(\Omega), \|\cdot\|_{H^1})$. The result is done in Theorem 1.12. The consideration of the space $H_0^1(\Omega)$ enables us to introduce $H^{-1}(\Omega)$ as the dual of $H_0^1(\Omega)$ (that is the set of linear forms defined on $H_0^1(\Omega)$) which is characterized by:

$$H^{-1}(\Omega) \stackrel{\text{def}}{=} (H_0^1(\Omega))' = \left\{ u \in \mathcal{D}'(\Omega) / u = f_0 + \sum_{i=1}^d \frac{\partial f_i}{\partial x_i}, f_0 \in L^2(\Omega), f_i \in L^2(\Omega), 1 \leq i \leq d \right\}$$

Remark 1.10. In $\mathbb{R}^d$, the Sobolev spaces can be defined thanks to the Fourier transform: for $s \in \mathbb{R}$,

$$H^s(\mathbb{R}^d) = \{u \in S'(\mathbb{R}^d) / (1 + |\xi|^2)^{s/2} \hat{u} \in L^2(\mathbb{R}^d)\}$$

This definition coincides with the previous ones and uses the concept of Fourier transform on the space of tempered distributions $S'(\mathbb{R}^d)$ which is the dual of the so-called Schwartz space $S(\mathbb{R}^d)$. The Schwartz space is the space of the functions of C^∞ which are rapidly decreasing at infinity.

Notions of trace operators and trace spaces

A boundary value problem requires the consideration of the restriction of functions to the boundary $\partial\Omega$ of the computational domain Ω . The mathematical justification of such consideration is given thanks to *trace operators*. The restrictions of functions which can be restricted to the boundary belong to *trace spaces*.

Theorem 1.11. Let Ω be a bounded open subset of $\mathbb{R}^d$, of class $\mathcal{C}^{0,1}$.

- i) $H_0^1(\Omega) = \{u \in H^1(\Omega) / \tilde{u} \in H^1(\mathbb{R}^d)\}$ where $\tilde{u}$ is the extension of u by 0 outside of Ω .
- ii) The mapping $\gamma_0 : u \mapsto u|_{\partial\Omega}$ is linear and continuous from $H^1(\Omega)$ to $L^2(\partial\Omega)$, and is called “trace operator”.

Theorem 1.12. Let Ω be bounded and of class $\mathcal{C}^{0,1}$. Then,

$$H_0^1(\Omega) = \{u \in H^1(\Omega) / \gamma_0 u = 0\} = \ker \gamma_0$$

Two trace spaces will be considered in the justification of the finite element method:

$$\begin{aligned} H^{1/2}(\partial\Omega) &= \{\underline{v} \in L^2(\partial\Omega) / \exists v \in H^1(\Omega) \text{ such that } \underline{v} = \gamma_0 v\} , \\ H^{3/2}(\partial\Omega) &= \{\underline{v} \in L^2(\partial\Omega) / \exists v \in H^2(\Omega) \text{ such that } \underline{v} = \gamma_0 v\} . \end{aligned}$$

1.2.2 Green's formulae

The well known one dimensional integration by parts is extended to higher dimensions by the so-called *Green's formula*:

Theorem 1.13 (Green's formula). *If $\Omega \subset \mathbb{R}^d$ is $\mathcal{C}^{0,1}$, then $\forall u, v \in H^1(\Omega)$, we have*

$$\int_{\Omega} \frac{\partial u}{\partial x_i} v = - \int_{\Omega} u \frac{\partial v}{\partial x_i} + \int_{\partial\Omega} \gamma_0 u \gamma_0 v \nu_i, \quad 1 \leq i \leq d, \quad (1.3)$$

where ν_i is the i -th component of the outgoing unit normal vector ν on $\partial\Omega$.

The demonstration of the formula is easy when $u, v \in \mathcal{C}^1(\Omega)$. The difficulty occurs in the consideration of Sobolev spaces. The result is extended to those spaces thanks to density theorems of functional analysis. In the sequel, we ignore the trace operator for readability and write

$$\int_{\Omega} \frac{\partial u}{\partial x_i}(x) v(x) dx = - \int_{\Omega} u(x) \frac{\partial v}{\partial x_i}(x) dx + \int_{\partial\Omega} u(\gamma) v(\gamma) \nu_i(\gamma) d\gamma, \quad 1 \leq i \leq d,$$

but the integral over $\partial\Omega$ of a function defined on Ω would have to be considered in terms of trace operator.

A first consequence of Green's formula is the following corollary that will be used for the first derivation of the 2D finite element method in this document:

Corollary 1.14. *If $\Omega \subset \mathbb{R}^d$ is $\mathcal{C}^{0,1}$, then $\forall u \in H^2(\Omega), \forall v \in H^1(\Omega)$, we have*

$$\int_{\Omega} \Delta u(x) v(x) dx = - \int_{\Omega} \nabla u(x) \cdot \nabla v(x) dx + \int_{\partial\Omega} \frac{\partial u}{\partial \nu}(\gamma) v(\gamma) d\gamma. \quad (1.4)$$

The generic formula associated with the generic problem defined in 1.1.2 is given by the corollary:

Corollary 1.15. *If $\Omega \subset \mathbb{R}^2$ is $\mathcal{C}^{0,1}$, $a_{\alpha\beta} \in H^1(\Omega)$, then $\forall u \in H^2(\Omega), \forall v \in H^1(\Omega)$, we have*

$$\begin{aligned} & \int_{\Omega} \sum_{\alpha, \beta=1}^2 \frac{\partial}{\partial x_{\beta}} \left(a_{\alpha\beta} \frac{\partial u}{\partial x_{\alpha}} \right) (x) v(x) dx \\ &= - \int_{\Omega} \sum_{\alpha, \beta=1}^2 a_{\alpha\beta}(x) \frac{\partial u}{\partial x_{\alpha}}(x) \frac{\partial v}{\partial x_{\beta}}(x) dx + \int_{\partial\Omega} \frac{\partial u}{\partial \nu_A}(\gamma) v(\gamma) d\gamma. \end{aligned} \quad (1.5)$$

1.2.3 From the PDE to the variational formulation

To derive the generic variational formulation, we multiply the generic PDE by a test function and integrate over the domain

$$- \int_{\Omega} \sum_{\alpha, \beta=1}^2 \frac{\partial}{\partial x_{\beta}} \left(a_{\alpha\beta}(x) \frac{\partial u}{\partial x_{\alpha}}(x) \right) v(x) dx + \int_{\Omega} a_{00}(x) u(x) v(x) dx = \int_{\Omega} f_{\Omega}(x) v(x) dx$$

Then, the global order of the derivatives is reduced thanks to the Green’s formula. The application of the formula (1.5) leads to

$$\begin{aligned} & \int_{\Omega} \sum_{\alpha, \beta=1}^2 a_{\alpha\beta}(x) \frac{\partial u}{\partial x_{\alpha}}(x) \frac{\partial v}{\partial x_{\beta}}(x) dx - \int_{\partial\Omega} \frac{\partial u}{\partial \nu_A}(\gamma) v(\gamma) d\gamma + \int_{\Omega} a_{00}(x) u(x) v(x) dx \\ &= \int_{\Omega} f_{\Omega}(x) v(x) dx \end{aligned} \quad (1.6)$$

Let us introduce some notations:

- ★ the unknown belongs to the space $W = H^1(\Omega)$;
- ★ the test function of the variational formulation is taken in W , satisfying $v|_{\Gamma_D} = 0$, $v \in V$ with

$$V = \{v \in W / v|_{\Gamma_D} = 0\};$$

- ★ for $u \in W$ and $v \in V$, we consider the bilinear form

$$\begin{aligned} a(u, v) = & \int_{\Omega} \left(\sum_{\alpha, \beta=1}^2 a_{\alpha\beta}(x) \frac{\partial u}{\partial x_{\alpha}}(x) \frac{\partial v}{\partial x_{\beta}}(x) + a_{00}(x) u(x) v(x) \right) dx \\ & + \int_{\Gamma_N} b_N(\gamma) u(\gamma) v(\gamma) d\gamma; \end{aligned} \quad (1.7)$$

and the linear form

$$f(v) = \int_{\Omega} f_{\Omega}(x) v(x) dx + \int_{\Gamma_N} g_N(\gamma) v(\gamma) d\gamma. \quad (1.8)$$

The combination of the boundary conditions, the definition of the spaces W and V , the definition of the forms a and f , together with the expression (1.6), leads to the variational form of the generic problem (1.2):

$$\left| \begin{array}{l} \text{Find } u \in W \text{ such that } u = u_D \text{ on } \Gamma_D \\ a(u, v) = f(v), \forall v \in V. \end{array} \right. \quad (1.9)$$

Regularity hypotheses on the functions (see Section 4.1) can ensure existence and uniqueness of the solution to (1.9). A fundamental lemma is done in Section 1.3 regarding this question.

1.2.4 The Dirichlet condition

The Dirichlet condition is not considered in the expression of the equation involved in the variational form. This condition only appears in the definition of the functional space. For the establishment of theoretical properties, it is usual to consider first the case of a homogeneous Dirichlet condition (i.e. $u_D = 0$), then to deal with the general Dirichlet condition by a change of unknown which links to the case of a homogeneous condition. Even in a practical point of view, it may be easier to implement the homogeneous Dirichlet condition ($u_D = 0$).

Let us consider the problem

$$\left| \begin{array}{l} \text{Find } u \in W \text{ such that } u = u_D \text{ on } \Gamma_D, \\ a(u, v) = f(v), \forall v \in V \end{array} \right. \quad (1.10)$$

where we suppose that there exists a function $U_D \in H^1(\Omega)$ such that $U_D|_{\Gamma_D} = u_D$. We then introduce a new unknown $U = u - U_D$ and the problem (1.10) becomes

$$\left| \begin{array}{l} \text{Find } U \in V \text{ such that } \forall v \in V \\ a(U, v) = f(v) - a(U_D, v), \end{array} \right. \quad (1.11)$$

Remark 1.16. *The function U_D is not unique. This means that the solution U depends on U_D but the solution $u = U + U_D$ does not depend on U_D .*

1.3 Wellposedness of variational formulations – Lax-Milgram Lemma

A fundamental result for the justification of the wellposedness of variational formulations is Lax-Milgram Lemma:

Lemma 1.17 (Lax-Milgram Lemma). *Let us consider a variational formulation*

$$\left| \begin{array}{l} \text{Find } u \in V \text{ such that } \forall v \in V \\ a(u, v) = f(v), \end{array} \right. \quad (1.12)$$

where

- ★ V is a Hilbert space,
- ★ a is a bilinear continuous form on $V \times V$,
- ★ f is a linear continuous form on V .

If the bilinear form a is coercive, that is

$$\exists \alpha > 0 / \forall u \in V, a(u, u) \geq \alpha \|u\|_V^2,$$

then, the variational formulation (1.12) has a unique solution.

The most difficult aspect of the theorem is the satisfaction of the coercivity of the bilinear form. For some problems, the Poincaré inequality may help:

Theorem 1.18 (Poincaré inequality). *Let Ω be a bounded subset of $\mathbb{R}^d$, of class $C^{0,1}$. There exists a constant $C_\Omega > 0$ depending only on Ω such that*

$$\forall u \in H_0^1(\Omega), \quad \|u\|_{L^2} \leq C_\Omega \|\nabla u\|_{L^2}.$$

Remark 1.19. *An equivalent result holds even if we force u to be 0 only on a part Γ_D of $\partial\Omega$ as soon as $\text{meas}(\Gamma_D) \neq 0$.*

Such results will lead to the following statements, for suitable hypotheses on Ω , f_Ω , g_N and u_D :

- ★ The following problem and its corresponding variational formulation have a unique solution:

$$\text{Find } u \in H^2(\Omega) \text{ such that } \begin{cases} -\Delta u + u = f_\Omega & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} = g_N & \text{on } \partial\Omega. \end{cases}$$

In this case, $V = H^1(\Omega)$, and the bilinear form is the inner product on $H^1(\Omega)$.

- ★ The following problem and its corresponding variational formulation do not have a unique solution:

$$\text{Find } u \in H^2(\Omega) \text{ such that } \begin{cases} -\Delta u = f_\Omega & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} = g_N & \text{on } \partial\Omega. \end{cases}$$

If we get a solution u , for any constant α , $u + \alpha$ is also a solution. The bilinear form of the variational formulation is not coercive.

- ★ The following problem and its corresponding variational formulation have a unique solution:

$$\text{Find } u \in H^2(\Omega) \text{ such that } \begin{cases} -\Delta u = f_\Omega & \text{in } \Omega, \\ u = u_D & \text{on } \partial\Omega. \end{cases}$$

The bilinear form of the variational formulation is coercive thanks to Poincaré inequality.

1.4 Galerkin method: discretization of variational formulations

In this section, we consider a general 2D configuration with homogeneous Dirichlet condition, based on a generic variational formulation

$$\begin{cases} \text{Find } u \in V \text{ such that } & \forall v \in V \\ & a(u, v) = f(v). \end{cases} \quad (1.13)$$

The functional space V , and the bilinear form a and the linear form f are supposed to satisfy some properties which ensure existence and uniqueness of the solution. For example, they satisfy the hypotheses of Lemma (1.17). Moreover, we consider that the space V contains the homogeneous Dirichlet condition on Γ_D : $\forall v \in V, v|_{\Gamma_D} = 0$. We derive formally a finite element approximation and the different aspects of its implementation. This section introduces the general concept of finite elements and indicates the usual strategy of application and implementation.

From the variational formulation, the first step consists in the choice of a partition of the computational domain Ω named a *mesh* or a *triangulation* when it consists of triangles, for the discretization of the geometry. For the sake of simplicity, we consider and describe here a triangulation $\mathcal{T}_h$ of Ω , a set of triangles K satisfying:

- i) $\bar{\Omega}_h = \bigcup_{K \in \mathcal{T}_h} K$ is a polygonal approximation of $\bar{\Omega}$, of boundary $\Gamma_{N,h} \cup \Gamma_{D,h}$ with $\Gamma_{N,h}$ (resp. $\Gamma_{D,h}$) approximation of Γ_N (resp. Γ_D);

- ii) $\forall K \in \mathcal{T}_h, \overset{\circ}{K} \neq \emptyset$;
- iii) $\forall K_1 \in \mathcal{T}_h, \forall K_2 \in \mathcal{T}_h, \overset{\circ}{K}_1 \cap \overset{\circ}{K}_2 = \emptyset$ or $\overset{\circ}{K}_1 = \overset{\circ}{K}_2$;
- iv) $\forall K'$ edge of $K_1 \in \mathcal{T}_h$, K' is either an edge of another triangle $K_2 \in \mathcal{T}_h$, or a part of $\Gamma_{D,h}$ or a part of $\Gamma_{N,h}$.

The parameter $h = \max_{K \in \mathcal{T}_h} \text{diam}(K)$ measures how fine the triangulation is. “diam” stands for “diameter”. For a triangle, the diameter is the length of the longest edge. The triangulation defines an approximation of the geometry when the domain Ω is not polygonal, otherwise it is possible to choose the triangulation such that $\Omega_h = \Omega$.

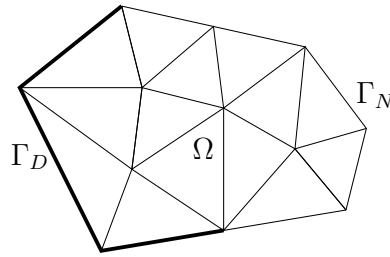


Figure 1.3: (generated with Fig4TeX [7]) Example of a triangulation of a polygonal domain Ω

The initial variational formulation (1.13) is now replaced with

$$\begin{cases} \text{Find } u \in \tilde{V} \text{ such that } & \forall v \in \tilde{V} \\ & a_h(u, v) = f_h(v), \end{cases} \quad (1.14)$$

where $\tilde{V}$, a_h and f_h are defined on Ω_h like V , a and f are defined on Ω (e.g., if $V = H^1(\Omega)$, then $\tilde{V} = H^1(\Omega_h)$).

A second step consists in the approximation of the space $\tilde{V}$ by a finite dimensional space V_h . Let us consider V_h defined by a basis $(\varphi_i)_{i=1,\dots,n}$ where n denotes the dimension of V_h . The functions $\varphi_i, i = 1, \dots, n$ are the so-called *basis functions*.

Replacing $\tilde{V}$ by V_h , the unknown u of (1.14) is approximated by $u_h = \sum_{j=1}^n u_j \varphi_j$, with $\{\varphi_i, i = 1, \dots, n\}$ a basis of V_h , and the equation in (1.14) requires only to be written for v in $\{\varphi_i, i = 1, \dots, n\}$. This leads to the system

$$\forall i = 1, \dots, n, \quad \sum_{j=1}^n u_j a_h(\varphi_j, \varphi_i) = f_h(\varphi_i) \quad (1.15)$$

where the unknown is the vector $(u_1, \dots, u_n) \in \mathbb{R}^n$. The system (1.15) can be written in matricial form:

$$AX = B \quad (1.16)$$

with the matrix $A \in \mathbb{R}^{n \times n}$ and the right hand side $B \in \mathbb{R}^n$ defined by

$$A_{ij} = a_h(\varphi_j, \varphi_i), \quad B_i = f_h(\varphi_i) \quad (1.17)$$

and the unknown in $\mathbb{R}^n$: $X = (u_1, \dots, u_n)^T$. This corresponds to a Galerkin formulation of the initial problem (1.13), a finite dimensional variational formulation.

Next step consists in specifying the choice of V_h or more exactly the choice of a set of basis functions $\{\varphi_i, i = 1, \dots, n\}$. Indeed, the finite element method is such a Galerkin method which answers this question with remarkable properties:

- ★ ease of calculation of the expressions $a_h(\varphi_j, \varphi_i)$ and $f_h(\varphi_i)$ which usually involve evaluations of integrals;
- ★ explicit expression of the basis $\{\varphi_i, i = 1, \dots, n\}$ of the functional space V_h ;
- ★ convenient properties of the matrix A for the resolution of the system;
- ★ convergence of the solution u_h to the linear system towards the solution to the variational formulation on Ω .

These aspects are carefully discussed in next chapter for a toy 2D problem and more generally considered in the following chapters.

Chapter 2

A complete 2D Lagrange Finite Element application

In this chapter, we propose to derive carefully the finite element approximation in the case of a 2D classical problem where the required mathematical properties are rather easy to check.

2.1 A toy 2D problem

We consider the following problem

$$\left\{ \begin{array}{l} \text{Find } u \in H^2(\Omega) \text{ such that} \\ -\Delta u + u = f_\Omega \quad \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} = g_N \quad \text{on } \Gamma_N \quad (\text{Neumann condition}), \\ u = u_D \quad \text{on } \Gamma_D \quad (\text{Dirichlet condition}), \end{array} \right. \quad (2.1)$$

where $\Omega \in \mathbb{R}^2$, Γ_N and Γ_D are defined like in Section 1.1, see Figure 1.2. If Ω is regular enough, considering $f_\Omega \in L^2(\Omega)$, $u_D \in H^{3/2}(\Gamma_D)$ and $g_N \in H^{1/2}(\Gamma_N)$, the problem is wellposed: the solution exists and is unique.

2.2 Variational formulation and wellposedness

To derive the variational form of (2.1), we first multiply the equation on Ω by a test function v satisfying $v|_{\Gamma_D} = 0$ and integrate over Ω :

$$-\int_{\Omega} \Delta u(x) v(x) dx + \int_{\Omega} u(x) v(x) dx = \int_{\Omega} f_\Omega(x) v(x) dx.$$

By considering the Green's formula (1.4), we obtain

$$\int_{\Omega} \nabla u(x) \cdot \nabla v(x) dx + \int_{\Omega} u(x) v(x) dx = \int_{\Omega} f_\Omega(x) v(x) dx + \int_{\partial\Omega} \frac{\partial u}{\partial \nu}(\gamma) v(\gamma) d\gamma.$$

The consideration of the boundary conditions (vanishing the test function on the Dirichlet boundary, using the Neumann condition) gives

$$\int_{\Omega} \nabla u(x) \cdot \nabla v(x) dx + \int_{\Omega} u(x) v(x) dx = \int_{\Omega} f_\Omega(x) v(x) dx + \int_{\Gamma_N} g_N(\gamma) v(\gamma) d\gamma.$$

To ensure the wellposedness of all the terms in the previous expression, the suitable functional space is $H^1(\Omega)$. Then, the variational form of Problem (2.1) is

$$\left| \begin{array}{l} \text{Find } u \in W \text{ such that } u|_{\Gamma_D} = u_D, \\ a(u, v) = f(v), \forall v \in V, \end{array} \right. \quad (2.2)$$

with

$$\begin{aligned} W &= H^1(\Omega), \quad V = \{v \in W / v|_{\Gamma_D} = 0\}, \\ a(u, v) &= \int_{\Omega} \nabla u(x) \cdot \nabla v(x) dx + \int_{\Omega} u(x)v(x) dx, \\ f(v) &= \int_{\Omega} f_{\Omega}(x)v(x) dx + \int_{\Gamma_N} g_N(\gamma)v(\gamma) d\gamma. \end{aligned} \quad (2.3)$$

The bilinear form a is indeed the usual inner product on $H^1(\Omega)$, and the linear form f is continuous due to the hypothesis on f_{Ω} and the trace theorems of functional analysis. The variational formulation satisfies the hypotheses of Lax-Milgram Lemma 1.17, the existence and uniqueness of the solution is ensured and the solution to (2.2) is the solution to (2.1). Existence and uniqueness of the solution to (2.2) can be checked with weaker hypotheses, $f_{\Omega} \in H^{-1}(\Omega)$, $u_D \in H^{1/2}(\Gamma_D)$ and $g_N \in H^{-1/2}(\Gamma_N)$, but the solution to (2.2) would not be a solution to (2.1) in a standard sense anymore. Such a solution would belong to $H^1(\Omega) \setminus H^2(\Omega)$ and would be said *weak solution to (2.1)* or *distributional solution*.

2.3 $\mathbb{P}_1$ Lagrange triangulation

For the mesh of the domain Ω , we choose a triangulation $\mathcal{T}_h$ which consists of $\mathbb{P}_1$ triangles. Each triangle is defined by its three vertices. We suppose that $\mathcal{T}_h$ satisfies the usual properties:

- i) $\bar{\Omega}_h = \bigcup_{K \in \mathcal{T}_h} K$ is a polygonal approximation of $\bar{\Omega}$, of boundary $\Gamma_{N,h} \cup \Gamma_{D,h}$ with $\Gamma_{N,h}$ (resp. $\Gamma_{D,h}$) approximation of Γ_N (resp. Γ_D);
- ii) $\forall K \in \mathcal{T}_h, \overset{\circ}{K} \neq \emptyset$;
- iii) $\forall K_1 \in \mathcal{T}_h, \forall K_2 \in \mathcal{T}_h, \overset{\circ}{K}_1 \cap \overset{\circ}{K}_2 = \emptyset$ or $\overset{\circ}{K}_1 = \overset{\circ}{K}_2$;
- iv) $\forall K'$ edge of $K_1 \in \mathcal{T}_h$, K' is either an edge of another triangle $K_2 \in \mathcal{T}_h$, or a part of $\Gamma_{D,h}$ or a part of $\Gamma_{N,h}$.

The parameter $h = \max_{K \in \mathcal{T}_h} \text{diam}(K)$ measures how fine the triangulation is. We refer to Figure 1.3 of Section 1.4 for an example of such a triangulation.

For the definition of the finite dimensional functional space, we introduce a $\mathbb{P}_1$ -piecewise interpolation of the unknown function based on the knowledge of the function at the vertices of the triangles. This corresponds to the so-called $\mathbb{P}_1$ *Lagrange finite elements*. The finite element related to a given triangle K is defined by :

- * the geometrical element, entirely described by the vertices a_1^K, a_2^K, a_3^K of the triangle K ;
- * the functional space on K : $\mathbb{P}_1(K)$ = space of polynomials of degree at most 1 with respect to all its variables considered all together;

- ★ the basis functions of $\mathbb{P}_1(K)$ associated with the nodes a_1^K, a_2^K, a_3^K : $\lambda_1^K, \lambda_2^K, \lambda_3^K$ of $\mathbb{P}_1(K)$ defined by $\lambda_n^K(a_\ell^K) = \delta_n^\ell$ for $n, \ell = 1, 2, 3$.

The basis functions satisfy the following properties

- ★ $(\lambda_1^K, \lambda_2^K, \lambda_3^K)$ is a basis of $\mathbb{P}_1(K)$;
- ★ If $p \in \mathbb{P}_1(K)$ satisfies $p(a_\ell^K) = \alpha_\ell$ for $\ell = 1, 2, 3$, then p is uniquely determined and $p = \sum_{\ell=1}^3 \alpha_\ell \lambda_\ell^K$;
- ★ $\lambda_1^K + \lambda_2^K + \lambda_3^K = 1$;
- ★ $x = \sum_{\ell=1}^3 a_\ell^K \lambda_\ell^K(x) \quad \forall x \in \mathbb{R}^2$.

The functions $\lambda_1^K, \lambda_2^K, \lambda_3^K$ are called barycentric coordinates with respect to the vertices a_1^K, a_2^K, a_3^K of the triangle K .

The definition of the basis functions is based on the association of each basis function with one point of K . These points which are associated with the basis functions are called the *nodes* of the finite element. In the case of $\mathbb{P}_1$ Lagrange finite elements, the nodes are the vertices but this is not the case for more general finite elements (see next chapter). The $\mathbb{P}_1$ Lagrange finite element is illustrated in Figure 2.1 where the points $n_\ell, \ell = 1, \dots, 3$, are the nodes of the finite element.

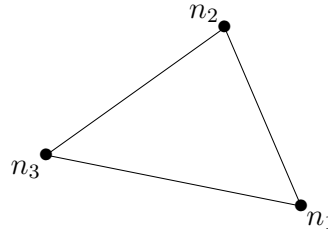


Figure 2.1: (generated with Fig4TeX [7]) Set of nodes of the $\mathbb{P}_1$ Lagrange finite element

2.4 Discrete space, numbering, basis functions

Let us denote by $\mathcal{N}^K$ the set of nodes of the element $K \in \mathcal{T}_h$, and $N^K = \text{card}(\mathcal{N}^K)$. In the current configuration of $\mathbb{P}_1$ finite elements, $N^K = 3$, and the following properties are satisfied: for any triangles K_1 and K_2 sharing an edge,

- ★ $\mathcal{N}^{K_1} \cap K_2 = \mathcal{N}^{K_2} \cap K_1$;
- ★ $\mathbb{P}_1(K_1)|_{K_1 \cap K_2} = \mathbb{P}_1(K_2)|_{K_1 \cap K_2}$.

We now introduce $\mathcal{M}_h$ the set of nodes of the triangulation $\mathcal{T}_h$ as the union of the nodes of the elements:

$$\mathcal{M}_h = \bigcup_{K \in \mathcal{T}_h} \mathcal{N}^K$$

and $M_h = \text{card}(\mathcal{M}_h)$. We introduce in the same way $\mathcal{N}_h$ the set of nodes of $\mathcal{M}_h$ which are not contained in Γ_D , and $N_h = \text{card}(\mathcal{N}_h)$. The set of elements which contain the node $n \in \mathcal{M}_h$ is

$$\mathcal{L}_h(n) = \{K \in \mathcal{T}_h / n \in \mathcal{N}^K\}.$$

These notations enable us to define the finite dimensional spaces

$$\begin{aligned} W_h &= \{v = (v_K)_{K \in \mathcal{T}_h} \in \prod_{K \in \mathcal{T}_h} \mathbb{P}_1(K) / \forall n \in \mathcal{M}_h, \forall K_1, K_2 \in \mathcal{L}_h(n), v_{K_1}(n) = v_{K_2}(n)\} \\ V_h &= \{v \in W_h / \forall n \in \mathcal{M}_h \cap \Gamma_D, \forall K \in \mathcal{L}_h(n), v_K(n) = 0\} \end{aligned}$$

For two adjacent elements K_1 and K_2 , the functions v_{K_1} and v_{K_2} coincide at the nodes of $K' = K_1 \cap K_2$. In the case of $\mathbb{P}_1$ -Lagrange finite elements, these objects coincide at the interfaces between two elements such that an object $v \in W_h$ is indeed a function in $\mathcal{C}^0(\Omega)$ and for any $K \in \mathcal{T}_h$, $v_K = v|_K$:

$$W_h = \{v \in \mathcal{C}^0(\Omega) / \forall K \in \mathcal{T}_h, v|_K \in \mathbb{P}_1(K)\},$$

However, it is important to mention that in a general configuration, the objects of W_h are not necessarily functions even if it is the case for any $\mathbb{P}_k$ -Lagrange finite elements, $k > 0$ (defined in next chapter).

A function $v \in W_h$ is then fully and uniquely determined by the knowledge of its values $\{v(n)\}_{n \in \mathcal{M}_h}$ at the nodes of $\mathcal{M}_h$, associated with $\mathcal{T}_h$, and called the *degrees of freedom* (d.o.f.) of W_h .

A crucial ingredient in the implementation of the finite element method is the transition from local numbering to global numbering. A function g^K is introduced to define this transition: the node n_i^K of K , $i = 1, \dots, N^K$, is the node n_I of $\mathcal{M}_h$, $I \in \{1, \dots, M_h\}$, identified by the relation $I = g^K(i)$. Moreover, the functions g^K , $K \in \mathcal{T}_h$, satisfy the property: $\forall n_I \in \mathcal{M}_h, \forall K_1 \in \mathcal{L}_h(n_I), \forall K_2 \in \mathcal{L}_h(n_I), \{(g^{K_1}(i) = g^{K_2}(j) = I) \iff (n_i^{K_1} = n_j^{K_2} = n_I)\}$.

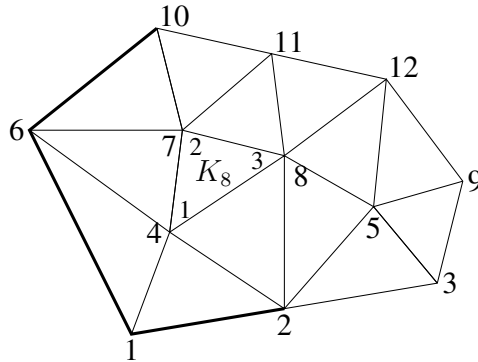


Figure 2.2: (generated with Fig4TeX [7]) Example of local and global numbering, $g^{K_8}(\{1,2,3\}) = \{4, 7, 8\}$

We now define the basis functions of the finite dimensional spaces. Let n_J be a node of $\mathcal{M}_h$, $J \in \{1, \dots, M_h\}$, we denote by w_J the basis function related to this node:

$$w_J \in W_h \quad \text{and} \quad w_J(n_I) = \delta_I^J, I = 1, \dots, M_h,$$

where δ_I^J is the Kronecker delta, equal to 1 if $I = J$ and 0 otherwise. One can check that

- ★ $\{w_J\}_{J=1,\dots,M_h}$ is a basis of W_h , and $\dim W_h = M_h = \text{card}(\mathcal{M}_h)$;
- ★ $\text{supp} w_J = \mathcal{L}_h(n_J)$;
- ★ $w_J|_K = w_j^K$ if there is $n_j^K \in \mathcal{N}^K$ such that $J = g^K(j)$, where w_j^K is the basis function associated with n_j^K on the element K . In our configuration of $\mathbb{P}_1$ Lagrange elements, $w_j^K = \lambda_j^K$.

2.5 Discrete problem and derivation of the linear system

The variational formulation (2.2) is replaced with the discrete variational form

$$\left| \begin{array}{l} \text{Find } u_h \in W_h \text{ such that } u_h|_{\Gamma_{D,h}} = u_{D_h}, \\ a_h(u_h, v_h) = f_h(v_h), \forall v_h \in V_h. \end{array} \right. \quad (2.4)$$

where

$$\begin{aligned} a_h(u_h, v_h) &= \int_{\Omega_h} \nabla u_h(x) \cdot \nabla v_h(x) dx + \int_{\Omega_h} u_h(x) v_h(x) dx, \\ f_h(v_h) &= \int_{\Omega_h} f_\Omega(x) v_h(x) dx + \int_{\Gamma_{N,h}} g_N(\gamma) v_h(\gamma) d\gamma, \\ u_{D_h} &\in W_h \text{ satisfying } u_{D_h}(n) = 0 \text{ for } n \in \mathcal{N}_h, \\ &\text{and } u_{D_h}(n) = u_D(n) \text{ for } n \in \mathcal{M}_h \setminus \mathcal{N}_h = \mathcal{M}_h \cap \Gamma_{D,h}. \end{aligned} \quad (2.5)$$

In order to simplify the notation, we consider that the nodes of $\mathcal{N}_h$ are numbered from 1 to N_h such that the nodes of $\mathcal{M}_h \cap \Gamma_D$ are numbered from $N_h + 1$ to M_h . Introducing the notation $u_J = u_h(n_J)$, $J = 1, \dots, M_h$, we can define u_h thanks to the basis of W_h :

$$u_h = \sum_{J=1}^{M_h} u_J w_J = \sum_{J=1}^{N_h} u_J w_J + \sum_{J=N_h+1}^{M_h} u_D(n_J) w_J,$$

The equation of (2.5) written for all $v_h \in V_h$ can equivalently be considered for the basis $\{w_I\}_{I=1,\dots,N_h}$ of V_h . The finite dimensional variational form is then equivalent to the problem

$$\left| \begin{array}{l} \text{Find } \{u_J\}_{J=1,\dots,N_h} \in \mathbb{R}^{N_h} \text{ such that } \forall I = 1, \dots, N_h, \\ \sum_{J=1}^{N_h} u_J a_h(w_J, w_I) = f_h(w_I) - \sum_{J=N_h+1}^{M_h} u_D(n_J) a_h(w_J, w_I) \end{array} \right. \quad (2.6)$$

which corresponds to the linear system

$$A_h X_h = F_h \quad (2.7)$$

with the matrix $A_h \in \mathbb{R}^{N_h \times N_h}$ and the right hand side $F_h \in \mathbb{R}^{N_h}$ defined by: $\forall I, J = 1, \dots, N_h$

$$(A_h)_{IJ} = a_h(w_J, w_I), \quad (F_h)_I = f_h(w_I) - \sum_{\tilde{I}=N_h+1}^{M_h} u_D(n_{\tilde{I}}) a_h(w_{\tilde{I}}, w_I) \quad (2.8)$$

and the unknown in $\mathbb{R}^{N_h}$: $X_h = (u_1, \dots, u_{N_h})^T$.

Due to the properties of the bilinear forms a and a_h and of the linear forms f and f_h of the variational formulations (2.2-2.3) and (2.4-2.5),

★ A_h is symmetric;

★ A_h is sparse:

$$\text{supp} w_I \cap \text{supp} w_J = \emptyset \implies (A_h)_{IJ} = 0,$$

which means

$$(A_h)_{IJ} \neq 0 \implies \mathcal{L}_h(n_I) \cap \mathcal{L}_h(n_J) \neq \emptyset.$$

The statement $(\mathcal{L}_h(n_I) \cap \mathcal{L}_h(n_J) \neq \emptyset)$ is equivalent to $(\exists K \in \mathcal{T}_h / n_I \in \mathcal{N}^K, n_J \in \mathcal{N}^K)$. Moreover, the matrix has a bandwidth the length of which can be determined by the formula

$$L = \max_{K \in \mathcal{T}_h} \max_{n_I, n_J \in K} |I - J|.$$

This indicates that the bandwidth of the matrix can be optimized by an appropriate numbering of the nodes of $\mathcal{T}_h$.

The first term of A_h , $\left(\int_{\Omega_h} \nabla w_J(x) \cdot \nabla w_I(x) dx \right)_{IJ}$, is known as the *stiffness matrix* and its second term, $\left(\int_{\Omega_h} w_J(x) w_I(x) dx \right)_{IJ}$, is known as the *mass matrix*.

2.6 Elementary matrices and right-hand-sides

In the finite element method, the matrix and right hand side of the linear system are computed through the consideration of *elementary matrices and right hand sides*. The elementary matrices and right hand sides are defined on the elements of the triangulation.

For $u_h, v_h \in W_h$, we introduce the notations: $\forall K \in \mathcal{T}_h$,

$$\begin{aligned} a_K(u_h, v_h) &= \int_K \nabla u_h(x) \cdot \nabla v_h(x) dx + \int_K u_h(x) v_h(x) dx, \\ f_K(v_h) &= \int_K f_\Omega(x) v_h(x) dx + \int_{K \cap \Gamma_{N,h}} g_N(\gamma) v_h(\gamma) d\gamma, \end{aligned} \tag{2.9}$$

Then, the matrix A_h and the right hand side F_h satisfy: $\forall I, J = 1, \dots, N_h$,

$$\begin{aligned} (A_h)_{I,J} &= \sum_{K \in \mathcal{L}_h(n_I) \cap \mathcal{L}_h(n_J)} a_K(w_J, w_I), \\ (F_h)_I &= \sum_{K \in \mathcal{L}_h(n_I)} f_K(w_I) - \sum_{\tilde{I}=N_h+1}^{M_h} \sum_{K \in \mathcal{L}_h(n_I) \cap \mathcal{L}_h(n_{\tilde{I}})} u_D(n_{\tilde{I}}) a_K(w_{\tilde{I}}, w_I). \end{aligned}$$

Combined with the definition of the function g^K , we get

$$\begin{aligned}
 (A_h)_{I,J} &= \sum_{K \in \mathcal{L}_h(n_I) \cap \mathcal{L}_h(n_J)} \sum_{i=1}^{N^K} \sum_{j=1}^{N^K} \delta_{g^K(i)}^I \delta_{g^K(j)}^J a_K(w_j^K, w_i^K), \\
 (F_h)_I &= \sum_{K \in \mathcal{L}_h(n_I)} \sum_{i=1}^{N^K} \delta_{g^K(i)}^I f_K(w_i^K) \\
 &\quad - \sum_{\tilde{I}=N_h+1}^{M_h} \sum_{K \in \mathcal{L}_h(n_I) \cap \mathcal{L}_h(n_{\tilde{I}})} \sum_{i=1}^{N^K} \sum_{j=1}^{N^K} \delta_{g^K(i)}^I \delta_{g^K(j)}^{\tilde{I}} u_D(n_{\tilde{I}}) a_K(w_j^K, w_i^K).
 \end{aligned} \tag{2.10}$$

where $\delta_{g^K(i)}^I$ is the Kronecker delta, equal to 1 if $I = g^K(i)$ and 0 otherwise. In this relation, indeed the summations “ $\sum_{i=1}^{N^K} \sum_{j=1}^{N^K}$ ” involve exactly one term.

The relation (2.10) indicates that the matrix and the right hand side of the linear system to be solved are entirely determined by the quantities $a_K(w_j^K, w_i^K)$, $f_K(w_i^K)$, for $K \in \mathcal{T}_h$, $i, j = 1, \dots, N^K$. These quantities correspond to the so-called *elementary matrices and right hand sides* associated with the triangulation $\mathcal{T}_h$:

Definition 2.1. Let K be an element of the triangulation $\mathcal{T}_h$, the elementary matrix and right hand side associated with K are the matrix A^K and the vector F^K defined by: $\forall i, j = 1, \dots, N^K$

$$A_{ij}^K = a_K(w_j^K, w_i^K), \quad F_i^K = f_K(w_i^K)$$

Thanks to this definition, the calculation of A_h and F_h reduces to:

- ★ the calculation of the elementary matrices and right hand sides;
- ★ the assembling of A_h and F_h from the elementary matrices and right hand sides.

The calculation of the elementary objects is done by a generic strategy which is defined through the consideration of a reference element. Essentially, the calculation of A^K and F^K consists of the explicit definition of the basis functions on K and quadrature rules for the numerical evaluation of the integrals. These objects are given on a reference element and the numerical approximation of A^K and F^K is obtained from this reference element by a change of variable. The assembling of A_h and F_h is done carefully in a manner which minimizes the algorithm cost of the operation.

2.7 The reference element

The reference triangle $\hat{K}$ is defined by the vertices $\hat{a}_1 = (1, 0)$, $\hat{a}_2 = (0, 1)$ and $\hat{a}_3 = (0, 0)$. The barycentric coordinates of $\hat{K}$ are defined by

$$\begin{cases} \hat{\lambda}_1(\hat{x}) = \hat{x}_1 \\ \hat{\lambda}_2(\hat{x}) = \hat{x}_2 \\ \hat{\lambda}_3(\hat{x}) = 1 - \hat{x}_1 - \hat{x}_2 \end{cases}$$

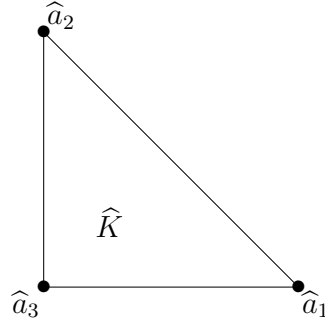


Figure 2.3: (generated with Fig4TeX [7]) The reference triangle

Any triangle K of $\mathcal{T}_h$ can be defined from the reference element. If we denote by a_ℓ^K , $\ell = 1, 2, 3$, the vertices of K , there exists a unique $\mathbb{P}_1$ function F_K defined on $\hat{K}$ such that $a_\ell^K = F_K(\hat{a}_\ell)$. F_K satisfies the following properties:

★ F_K is invertible for any non degenerate triangle K ;

★ $F_K(\hat{x}) = x = \sum_{\ell=1}^3 a_\ell^K \hat{\lambda}_\ell(\hat{x})$;

★ $\exists B_K \in \mathbb{R}^{2 \times 2}, b_K \in \mathbb{R}^2$ such that

$$F_K \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 \end{pmatrix} = B_K \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 \end{pmatrix} + b_K \quad (2.11)$$

★ B_K and b_K are explicitly defined by

$$B_K = \begin{pmatrix} a_{1,1}^K - a_{3,1}^K & a_{2,1}^K - a_{3,1}^K \\ a_{1,2}^K - a_{3,2}^K & a_{2,2}^K - a_{3,2}^K \end{pmatrix} \quad \text{and} \quad b_K = \begin{pmatrix} a_{3,1}^K \\ a_{3,2}^K \end{pmatrix}.$$

where $\{a_{\ell,j}^K\}_{j=1,2}$ are the components of a_ℓ^K .

In the same way, we define the reference segment $\hat{K}' = [0, 1]$ which is used for the calculation of the integrals on edges of the triangles which interact with the Neumann condition. A generic edge K' is obtained from the reference element thanks to the function

$$F_{K'}(\hat{\gamma}) = \gamma = (1 - \hat{\gamma})a_1^{K'} + \hat{\gamma}a_2^{K'}$$

where $\{a_\ell^{K'}\}_{\ell=1,2}$ are the extremities of the edge K' .

2.8 Elementary objects and reference element

Thanks to the reference element, the calculation of the elementary matrices and right hand sides is done through the change of variable defined by the function F_K :

$$\begin{aligned}
 a_K(w_j^K, w_i^K) &= \int_{\hat{K}} \nabla w_j^K(F_K(\hat{x})) \cdot \nabla w_i^K(F_K(\hat{x})) |\det JF_K(\hat{x})| d\hat{x} \\
 &\quad + \int_{\hat{K}} w_j^K(F_K(\hat{x})) w_i^K(F_K(\hat{x})) |\det JF_K(\hat{x})| d\hat{x}, \\
 f_K(w_i^K) &= \int_{\hat{K}} f_\Omega(F_K(\hat{x})) w_i^K(F_K(\hat{x})) |\det JF_K(\hat{x})| d\hat{x} \\
 &\quad + \sum_{K' \subset \partial K \cap \Gamma_{N,h}} \int_{\hat{K}'} g_N(F_{K'}(\hat{\gamma})) w_i^K(F_{K'}(\hat{\gamma})) |\det JF_{K'}(\hat{\gamma})| d\hat{\gamma},
 \end{aligned} \tag{2.12}$$

where $\det JF_K$ and $\det JF_{K'}$ are the determinants of the Jacobian matrices of F_K and $F_{K'}$.

We can check the remarkable properties:

- ★ For $x \in K$ and $\hat{x} \in \hat{K}$,

$$\hat{w}_i(\hat{x}) = w_i^K \circ F_K(\hat{x}), \quad w_i^K(x) = \hat{w}_i \circ F_K^{-1}(x)$$

- ★ For $\alpha = 1, 2$ and $x \in K$,

$$\begin{aligned}
 \frac{\partial w_i^K}{\partial x_\alpha}(x) &= \frac{\partial \hat{w}_i}{\partial \hat{x}_1} \circ F_K^{-1}(x) \frac{\partial (F_K^{-1})_1}{\partial x_\alpha}(x) + \frac{\partial \hat{w}_i}{\partial \hat{x}_2} \circ F_K^{-1}(x) \frac{\partial (F_K^{-1})_2}{\partial x_\alpha}(x) \\
 &= (\widehat{\text{grad}} \hat{w}_i \circ F_K^{-1}(x))^T (JF_K^{-1}(x))_\alpha.
 \end{aligned}$$

- ★ In the current case of $\mathbb{P}_1$ Lagrange finite elements, with $\mathbb{P}_1$ triangles, the Jacobian matrix JF_K is constant with respect to $\hat{x}$ and equal to B_K .

These statements show that the elementary matrices and right hand sides are entirely determined by the knowledge of:

- ★ F_K for each element K (i.e. the vertices of K and the basis functions $\{\hat{\lambda}_\ell\}_{\ell=1,\dots,3}$, on $\hat{K}$);
- ★ the finite element basis functions on $\hat{K}$, $\hat{w}_\ell$, $\ell = 1, \dots, 3$;
- ★ an integration strategy on the reference element $\hat{K}$.

In the current configuration of $\mathbb{P}_1$ Lagrange finite elements combined with a $\mathbb{P}_1$ Lagrange interpolation of the geometry (F_K affine), we have $\hat{\lambda}_\ell = \hat{w}_\ell$ but this is not necessary.

2.9 Quadrature rules

It is usually impossible to evaluate exactly some of the integrals involved in the definition of the elementary objects $a_K(w_j^K, w_i^K)$ or $f_K(w_i^K)$. Numerically, the evaluation is operated using quadrature rules. The following nodal rules are standard rules for the reference triangle or reference segment:

★ Quadrature rule on triangle, exact for $\mathbb{P}_1$ polynomials:

$$\int_{\hat{K}} \varphi(\hat{x}) d\hat{x} \approx \frac{S_{\hat{K}}}{3} (\varphi(\hat{a}_1) + \varphi(\hat{a}_2) + \varphi(\hat{a}_3)).$$

★ Quadrature rule on triangle, exact for $\mathbb{P}_2$ polynomials:

$$\int_{\hat{K}} \varphi(\hat{x}) d\hat{x} \approx \frac{S_{\hat{K}}}{3} (\varphi(\hat{a}_4) + \varphi(\hat{a}_5) + \varphi(\hat{a}_6)).$$

★ Quadrature rule on segment, exact for $\mathbb{P}_1$ polynomials:

$$\int_{\hat{K}'} \varphi(\hat{\gamma}) d\hat{\gamma} \approx \frac{L_{\hat{K}'}}{2} (\varphi(\hat{a}'_1) + \varphi(\hat{a}'_2)).$$

★ Quadrature rule on segment, exact for $\mathbb{P}_2$ polynomials:

$$\int_{\hat{K}'} \varphi(\hat{\gamma}) d\hat{\gamma} \approx \frac{L_{\hat{K}'}}{6} (\varphi(\hat{a}'_1) + 4\varphi(\frac{\hat{a}'_1 + \hat{a}'_2}{2}) + \varphi(\hat{a}'_2)).$$

where $S_{\hat{K}}$ and $L_{\hat{K}'}$ are respectively the surface of $\hat{K}$ (1/2 for the reference triangle) and length of $\hat{K}'$ (= 1 for the reference segment).

2.10 Assembling procedure

The efficient procedure consists in considering the elements of $\mathcal{T}_h$ one by one: for each element, the contributions of its elementary objects are distributed to the corresponding matrix or right hand side element. The correspondence is specified by the function g^K .

A simple version of the procedure is written below, with no detail on the storage consideration. Indeed, due to the sparsity of the matrix, storage preoccupation is crucial and carefully answered in next chapter.

- Initialization of A_h and F_h to 0.
- For each element K :
 - For each couple of nodes (i, j) of K :
 - If $n_i^K \notin \Gamma_D, n_j^K \notin \Gamma_D$;
 - $I = g^K(i); J = g^K(j)$;
 - $(A_h)_{IJ} \leftarrow (A_h)_{IJ} + A_{ij}^K$;
 - End if;
 - If $n_i^K \notin \Gamma_D, n_j^K \in \Gamma_D$;
 - $I = g^K(i); J = g^K(j)$;
 - $(F_h)_I \leftarrow (F_h)_I - u_D(n_J) A_{ij}^K$;
 - End if;
 - End for (i, j) ;
 - For each i , node of K such that $n_i^K \notin \Gamma_D$:
 - $I = g^K(i)$;
 - $(F_h)_I \leftarrow (F_h)_I + F_i^K$;
 - End for i ;
- End for K ;